

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-a27b88e9f090807f4.jsonl`  
- SHA-256: `53e0881e02d78b4cf874628737a2e08651598f54039ce9038cb65898ea7543f5`  
- Source modified: `2026-07-03T16:45:34+00:00`  
- Imported at: `2026-07-05T16:48:26+00:00`  
- project: `subagents`  
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`
```

Transcript

1. user (2026-07-03T16:45:09.552Z)

You are a code-review VERIFIER. Repo: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer, branch codex/alpha-readiness-plan.

CANDIDATE: docs/ALPHA_LAUNCH_READINESS.md §2 "Decisions locked" table presents five decisions D1?D5 as locked. D1, D2, D5 cite owner-dated sources (NEXT_STEPS NO-GO banner 2026-06-30, P0 owner-priority item, khelsutra lifecycle doc). But D3 ("The 15-video corpus is now a measurement asset, not a free launch pass" ? source: GOLDEN_VIDEOS.md + "local eval runs, 2026-07-03") and D4 ("Do not promote served Gate A by default from the current n=15 evidence" ? source: "backend.eval.serve_contrast run, 2026-07-03") are sourced only from the doc author's own analysis runs on 2026-07-03 ? no owner sign-off, no ADR. Claim: labeling analyst-derived conclusions as "Decisions locked" overstates their authority in a repo whose convention is (a) an ADR log at docs/archives/decisions/DECISIONS.md for ratified decisions and (b) pervasive owner-gating language; a future reader will treat D3/D4 as owner-ratified when they are not.

Verify: (1) Read docs/ALPHA_LAUNCH_READINESS.md §2 exactly ? how are D3/D4 sourced vs D1/D2/D5 (quote the Source cells)? (2) Read docs/PROJECT_DOC_TEMPLATE.md ? what does it say a "decisions locked" section is for, and does it require owner ratification or an ADR? (3) Skim docs/archives/decisions/DECISIONS.md headers and docs/DOC_STATUS.md to see how past decisions (e.g. D7?D17 / ADR-014) were ratified ? is there an explicit convention that locked decisions need owner/ADR ratification, or do tracked docs routinely lock analysis-derived decisions locally? Decide whether the doc misrepresents D3/D4's authority or whether the template permits doc-local decisions and the sources are honestly labeled.

Return exactly one verdict: CONFIRMED (convention requires owner/ADR ratification and D3/D4 lack it ? quote the convention), PLAUSIBLE (no hard rule, but D3/D4's provisional nature is worth labeling given the repo's honesty conventions), or REFUTED (template permits doc-local locked decisions and the sourcing is already honest ? quote it). Final message: verdict word, 2-5 sentences of evidence with quotes, one-line suggested fix if CONFIRMED/PLAUSIBLE.

2. assistant (2026-07-03T16:45:12.534Z)

I'll systematically verify this claim by reading the three key documents and checking the decision-tracking conventions.

3. user (2026-07-03T16:45:12.717Z)

```
1      # Alpha Launch Readiness Plan  
2  
3      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**  
2026-07-03; **Last updated:** 2026-07-03  
4      > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`  
5      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed  
in `docs/README.md` and `docs/DOC_STATUS.md`.  
6      > **Relation to existing docs:** consolidates the launch-relevant pending items from  
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,
```

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`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.
7      > **Honesty note:** `[verified]` means checked against the local synced repos on
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
legal, financial, or launch approval.
8
9      ---
10
11     ## 0. TL;DR
12
13     The corpus is now large enough to move from "grow labels first" into alpha-readiness
work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden
manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before
manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion
schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed
before it can gate anything.
14
15     The first controlled alpha should be a hosted, authenticated product loop: upload a
video, process it, select rallies, compile, and download a reel. The owned-model/commercial
rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the
WASB checkpoint provenance is resolved.
16
17     ## 1. Problem & goal
18
19     The previous launch blockers were spread across roadmap docs and mixed together three
different questions:
20
21     - **Can an invited tester use the product without SSH?**
22     - **Can the 15-video golden corpus now support honest quality gates?**
23     - **Can we claim or ship an owned/commercial rally detector?**
24
25     This doc separates those into executable workstreams. The goal is to reach an invite-
only alpha where the product is usable end-to-end, the quality claims are honest and
reproducible, and the remaining model/legal gates are explicit rather than accidentally implied
by launch activity.
26
27     ## 2. Decisions locked
28
29     | # | Decision | Source / date | Implication |
30     |---|---|---|---|
31     | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
32     | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
33     | D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
completeness, baseline refresh, and gate ratification come before quality claims. |
34     | D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-
windowing safety; keep it as an opt-in/analysis lever. |
35     | D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens
D13.** | sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha
can use manual/offline corpus promotion; automated flywheel follows after serving and gates are
stable. |
36
37     ## 3. Verified snapshot, 2026-07-03
38
39     All five Git checkouts were fast-forwarded before analysis:
40
41     | Repo | Branch | Synced HEAD |
42     |---|---|---|
43     | `badminton-highlight-indexer` | `master` | `e4914d1` |
44     | `khelsutra` | `master` | `b0f70a5` |
45     | `rally-annotator` | `main` | `ef9e088` |

```

```

46 | `sports-data-collector` | `master` | `bfc014f` |
47 | `sports-obsreport` | `main` | `a0c7252` |
48
49 Corpus/eval facts:
50
51 - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
`output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
52 - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
53 - `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
54 - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only
6 videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
55 - `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
56 - `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
57 - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
58 - `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity
group` currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
59
60 ## 4. Launch strategy
61
62 ### 4.1 Alpha lane: make the product usable
63
64 The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most
reliable serving path.
65
66 ### 4.2 Quality lane: make the 15-video corpus gate-ready
67
68 The quality lane is successful when the 15-video corpus is portable, reproducible,
feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
69
70 ### 4.3 Flywheel lane: make tester feedback useful
71
72 The flywheel lane is successful when user corrections can flow into the collector/vault
path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning,
but it should not block the first controlled alpha if manual promotion can bridge the gap.
73
74 ## 5. Risks and guardrails
75
76 | Risk | Why it matters | Guardrail |
77 |---|---|---|
78 | Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
79 | Stale local manifests create false confidence | Manifest-driven tools skipped or
failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke
check |
80 | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still
n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
81 | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly
reported stale baselines | Refresh baselines only after reviewer sign-off and record the
command/output |
82 | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but
can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha
lane is already stable |

```

83 | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained
model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product
launch tasks |

84

85 ## 6. Honest limits

86

87 Completing this doc's alpha lane does ****not**** prove the owned model is ready, that WASB
weights are commercially clean, that the product can serve unbounded public traffic, that multi-
sport is ready, or that every tester correction automatically improves the model.

88

89 Completing the quality lane does ****not**** by itself deploy a service. It only makes the
measurement surface honest enough for a launch decision.

90

91 ## 7. Deliverables & progress tracker

92

93 Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
row explicitly names a cross-repo batch.

94

ID	Deliverable	Repo / owner	Depends on	Gated?	Status	PR
A0	Create this alpha readiness tracker and index it in docs	`badminton-highlight-indexer`	-	No	IN PROGRESS	#451
A1	Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified	`khesutra` + `badminton-highlight-indexer`	A0	Owner spend / CF config	TODO	-
A2	Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing	`badminton-highlight-indexer`	A0	No	TODO	-
A3	Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus	`sports-data-collector` + vault	A2	Owner upload/storage scope	TODO	-
A4	Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed	`badminton-highlight-indexer`	A2	Reviewer sign-off	TODO	-
A5	Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6` floor for future Gen-0 comparisons	`badminton-highlight-indexer`	A2	No	TODO	-
A6	Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus	`badminton-highlight-indexer`	A2, A4	No	TODO	-
A7	Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15	`badminton-highlight-indexer`	A2	GPU/data availability	TODO	-
A8	Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips	`badminton-highlight-indexer`	A6, A7	Reviewer sign-off	TODO	-
A9	Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict	`badminton-highlight-indexer`	A5, A8	Owner approval for M0.4 scope	TODO	-
A10	Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed	Owner + `badminton-highlight-indexer`	A4, A5, A8	Owner decision	GATED	-
A11	Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail	Owner/legal	A10	Legal/provenance	GATED	-
A12	Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train reeval` job, and delta-F1 surface	`khesutra` + `badminton-highlight-indexer`	A1, A3	Owner D13 reopen	GATED	-
A13	Alpha launch review: runbook, privacy/consent copy, known limits, rollback/disable steps, and owner GO/NO-GO	Cross-repo	A1, A4, A10, A11	Owner launch decision	TODO	-

111

112 ## 8. Concrete next PR sequence

113

1. ****Docs PR:**** land this tracker.
2. ****Corpus portability PR:**** rebase/portable manifest tooling plus missing-file smoke

```

test.
116     3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download
verification and runbook updates.
117     4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor
refresh.
118     5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
119     6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy
limits, and launch checklist.
120
121     ## 9. Open questions
122
123     1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
124     2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
125     3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains
unresolved?
126     4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
127     5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every
clip, or should the eval loaders support mixed schemas with explicit capability tags?
128
129     ## 10. Definition of done
130
131     - [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without
SSH.
132     - [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
133     - [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
134     - [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
135     - [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
136     - [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
137     - [ ] Ship-gate target and allowed alpha claims are recorded.
138     - [ ] WASB C3 status is recorded and reflected in alpha messaging.
139     - [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for
alpha-plus.
140     - [ ] Owner records GO/NO-GO for the first alpha wave.
141
142     ## 11. References
143
144     Internal docs:
145
146     - `docs/NEXT_STEPS.md`
147     - `docs/ROADMAP.md`
148     - `docs/data_pipeline/GOLDEN_VIDEOS.md`
149     - `docs/R2_EVAL_METRIC_DESIGN.md`
150     - `docs/RALLY_DETECTION_FIXES_PLAN.md`
151     - `docs/COMPUTE_DECOUPLED_SERVING/README.md`
152     - sibling `khehsutra/NEXT_STEPS.md`
153     - sibling `khehsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
154     - sibling `khehsutra/docs/PER_RALLY_SELECTION.md`
155     - sibling `sports-data-collector/data/golden_source_videos.json`
156
157     Local verification commands used for the 2026-07-03 snapshot:
158
159     ```powershell
160     python -m backend.eval.fusion_compare --features-dir output
161     python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
162     python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
163     python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir
output --no-report

```

```

164     python -m backend.eval.ablation --features-dir output --granularity group
165     ``
166
167     ---
168
169     ### Changelog
170
171     - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
172

```

4. user (2026-07-03T16:45:12.841Z)

```

1     <!--
2     PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3
4     A "tracked project doc" is how this repo carries substantial design/project work: it can
be
5     ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6     archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7     docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9     Rules of thumb:
10    - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11    - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12      or [design] (proposed). The doc reflects real shipped state, never aspirational.
13    - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
14    - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15      research- or security/privacy-heavy work). Delete sections that do not apply.
16    - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17    -->
18
19    # <Project / Plan Title>
20
21    > **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` .
**Created:** `<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
22    > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23    > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
24    > **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
25    > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
26
27    ---
28
29    ## 0. TL;DR
30    <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32    ## 1. Problem & goal
33    <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35    ## 2. Decisions locked
36    <Table ? capture answered questions so they are not relitigated.>
37
38    | # | Decision | Source / date | Implication |
39    |---|-----|-----|-----|
40    | D1 | ? | ? | ? |

```

```

41
42     ## 3. Foundation ? research / prior art *(OPTIONAL)*
43     <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45     ## 4. Design / architecture
46     <The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing
modules/code this builds on so it is clear what is new vs reused.>
47
48     ## 5. Threat model / risk table *(OPTIONAL ? security / privacy / compliance work)*
49     <Who/what is protected vs NOT; residual risks stated plainly.>
50
51     ## 6. Honest limits ? what this does NOT do
52     <The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>
53
54     ## 7. Deliverables & progress tracker    ? **source of truth**
55     Legend: ? Todo ? In progress ? Done ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
56
57     | ID | Deliverable | Depends on | Gated? | Status | PR |
58     |----|-----|-----|-----|-----|----|
59     | P0 | ? | ? | No | ? | ? |
60
61     ## 8. Open questions ? owner / external *(split blocking-gates from nice-to-knows; name
who decides)*
62     1. ?
63
64     ## 9. Definition of done
65     <Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
66     - [ ] ?
67
68     ## 10. References
69     **Internal code anchors `[verified]`:** `<paths>`. **Internal docs:** `<docs>`.
**External:** `<cited sources>`.
70
71     ---
72
73     ### Changelog
74     - `<YYYY-MM-DD>` ? `<what changed>`.
75

```

5. user (2026-07-03T16:45:13.259Z)

```

1     # Documentation Status & Archival Register
2
3     > **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-
diligence**. **Last full review:** 2026-07-02.
4     > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a
milestone / delivered / archived), and re-review at each clean-slate.
5     > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update
it as work finishes** to prove we're done, and (c) **never archive a foundational doc without
honestly updating it first** (owner directive, 2026-06-21).
6
7     ---
8
9     ## 0. How to use this
10    - **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention.
§3 is the full register.
11    - **Marking something complete?** Update the doc's own **§7 progress tracker** (the
source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2
archival checklist.
12    - **This register does NOT duplicate completeness detail** ? it indexes lifecycle +
archive-readiness and points to each tracked doc's §7.
13

```

```

14     ## 1. Lifecycle legend
15     `DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent
reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal
record of a run/launch), **CONVENTION** (a standing rule/guide).
16
17     ## 2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git
mv`)*
18     Before a foundational/tracked doc moves to `docs/archives/`:
19     1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts
`[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it
first.)
20     2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped,
where (PRs, paths, artifacts), and what was **deferred / carried forward**.
21     3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`,
`docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
22     4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the
evolution stays traceable.
23     5. **git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state
ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this
in the doc's header).
24     6. **Record the move** ? flip the row here to $3e + a line in memory `current-state`.
25
26     > **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5
delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward
WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014
lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.
27
28     ## 3. The register
29
30     ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's $7 is the
completeness source-of-truth)
31     | Doc | Status | Archive-ready? | Next action |
32     |---|---|---|---|
33     | `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created
2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-
model/commercial rally-detection gates | No | Execute $7 in order: corpus portability + n=15
baselines, hosted alpha serving, ablation/fusion reruns, then launch G0/NO-G0 |
34     | `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY
SHIPPED** (#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0
weights pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual:
counsel/prices/OAuth/real-run within the D17 cap) |
35     | `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** ($7) | No | Drives the
R-series; complete when $7.1 single quality number is attributed |
36     | `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval;
ablation pending) | Promote-or-reject toIF1 as the gate (ablation) ? then archive with the
R-cluster |
37     | `R1_MILESTONE_LAUNCH_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the
R-cluster) | Keep with R-series for narrative; archive when the track wraps |
38     | `RALLY_DETECTION_GAP_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of
the R/quality cluster |
39     | `RALLY_DETECTION_FIXES_PLAN.md` | **DONE** ? findings A/B/C resolved; P2b promoted
`inactivity_temporal_density` default-ON (#396) | With the R-cluster (deferred per its header;
gate = R2 toIF1 ablation + $7.1 number) | Archive with the R-series when the track wraps |
40     | `RALLY_DETECTION_CODE_REVIEW.md` | Terminal snapshot ? findings A (#394), B (#394 ?
promoted #396), C (`sdc#50`) all resolved | With the R-cluster | Archive with the R-series |
41     | `PER_VIDEO_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile
(finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates
with M8's v6 cost-ledger ? version bump authored once) |
42     | `MULTI_SIGNAL_FUSION_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion,
audio) ? needs GPU golden features |
43     | `EXPERIMENT_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the
quality track |
44     | `PROMINENT_COURT_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track
selection design |
45     | `EXECUTION_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud

```



```

Run dispatch |
46      | `GOLDEN_REGRESSION_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval
fixtures |
47      | **Data layer** ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` .
`GOLDEN_SET_PHASE1_VIDEOS` . `GOLDEN_VIDEOS` . `VIDEO_RECORDING_GUIDELINES` .
`GOLDEN_DATA_SHARING`) | LIVING (data, **no-ML**) | n/a | Relocated 2026-06-21 to the isolated
data pillar; tracked via `data_pipeline/README.md`. Roora *vendor* docs archived ?
`archives/roora-vendor/`. |
48      | `archives/past_projects/AUDIT_REPORT_khelsutra-
guru_2026-07-02-COMPLETED-2026-07-02.md` **(umbrella)** | **? COMPLETE / ARCHIVED (2026-07-02)**
? the five-repo code audit + folded-in *Remediation tracker* (P0?P28 all closed) + absorbed
prior audits. All non-owner-gated residuals landed; owner-gated/future items filed as issues
(#443/#444/#445, + #301/#327/#332/#384). | Archived | Owner confirmed archival 2026-07-02; moved
to `archives/past_projects/` per §2. Residuals now live as GitHub issues, not in this doc. |
49      | `DECISION_SNAPSHOT_REGRESSION.md` | **DRAFT ? decisions LOCKED** (D1?D11 locked
2026-06-25, #383). Zero code shipped for P1?P7. | No | Start P1+P2 (schema + stitch-plan
factor); de-risk the P10 main.py split |
50      | `DESKTOP_APP_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-
feasibility spike; relates to truly-local profile |
51      | `OWNED_MODEL_IMPLEMENTATION_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative
roadmap) | Gen-0?Gen-2; the owned-model track |
52      | `PERSONALIZATION_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB
v5 store landed) |
53      | `UI_ALPHA_EXECUTION_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |
54      | `I18N_PLAN.md` | Design (P0); **in implementation** (adapted) | No | Executed in
`khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-
gated |
55      | `POST_166...RISK_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |
56
57      ### 3b. Foundational / strategy references (LIVING ? archive only if **superseded**, not
when "done")
58      | Doc | Status | Note |
59      |---|---|---|
60      | `PLATFORM_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend`
registries ? realized by COMPUTE_DECOUPLED_SERVING |
61      | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |
62      | `VIDEO_LOCALITY_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |
63      | `HOSTING_PLAN.md` | PROPOSAL | **? partly overtaken by ADR-013 + the shipped site ?
verify (finding #5)** |
64      | `UI_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half **in implementation** ?
`khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc)
|
65      | `STORAGE_SHARING_MODEL.md` | DRAFT (co-design pending) | ? |
66      | `ANALYZERS_AND_RECONCILERS.md` | DESIGN (owner-gated) | ? |
67      | `ROADMAP.md` / `CODE_MAP.md` / `HOW_RALLY_DETECTION_WORKS.md` /
`KHELSUTRA_PRODUCT_OVERVIEW.md` | LIVING references | Keep current; not archive targets |
68
69      ### 3c. Reports & terminal records (archive candidates once their track wraps)
70      | Doc | Status | Note |
71      |---|---|---|
72      | `REAL_FOOTAGE_VALIDATION_REPORT.md` | validation complete | Motivates R2; archive with
the R-cluster |
73      | `RALLY_DETECTION_QUALITY_REPORT.md` | report | Quality snapshot; part of R-cluster |
74      | `QUALITY_ITERATIONS.md` | LIVING empirical log | **Keep** ? the paper-backbone
ablation log ([[paper-coauthor-aim]]) |
75      | `CODE_AUDIT_AND_TEST_HARDENING` . `HARDENING_LOOP_HANDOFF` . `DEFERRED_HARDENING_PLAN`
| COMPLETED ? **finish-archived 2026-07-02** (top-level stubs removed; absorbed into the
umbrella's *Prior audits*) | ? Full copies under `docs/archives/*-COMPLETED-2026-06-14.md`;
inbound links repointed (finding #4 resolved) |
76
77      ### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)
78      `PROJECT_DOC_TEMPLATE.md` . `NEXT_STEPS.md` (? refreshed 2026-06-21) . `BACKLOG.md` .
`EVALUATION.md` . `ABLATION_LAYER.md` . `SETUP_NEW_MACHINE.md` . **`CREATING_REELS.md`** (? end-
user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) .
**`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`** (? operator runbook ? `gs://` videos ? reels on L4

```

Cloud Run; verified e2e 2026-06-23) · `TRACKNET\_WSL\_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data\_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

79

80 ### 3e. Already archived (`docs/archives/`)

81 `past\_projects/DECOUPLED\_COMPUTE/` (? 2026-06-21, #270) · `past\_projects/low-regret-moves/` · **field-pmf/`\*\* (field-ops + customer-feedback, 2026-06-21) · \*\*`roora-vendor/`\*\* (annotation-vendor engagement, 2026-06-21) · `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL_READINESS_REVIEW.md` · `MONETIZATION_AUDIT.md` · `TEST_RUN_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/` ·

82 `past\_projects/CODE\_CLEANUP\_AUDIT\_2026-06-24-SUPERSEDED-2026-07-02.md` (absorbed ? umbrella) · `past\_projects/KHELSUTRA\_GURU\_HEALTH\_REMEDIATION\_PLAN-FOLDED-2026-07-02.md` (folded ? umbrella).

83 ### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

84 **ADR-001 ? ADR-014** (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014**.

85 ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).

86 ## 4. Open due-diligence findings / actions (2026-06-21 review)

87

88 > **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data layer** carved into `docs/data\_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data\_pipeline` one sets the "no segmentation/stitching" boundary).

89 1. **? RESOLVED (2026-06-21) ? `PER\_VIDEO\_WORKSPACE.md` supersedes `PER\_VIDEO\_OUTPUT\_LAYOUT.md`.** Folded forward the `video\_slug` naming option (workspace §8.5) + the `output/GX010125\_run/` manual-prototype precedent; predecessor archived ?

90 2. **? RESOLVED (2026-06-21) ? `NEXT\_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**;

91 3. **R-series / quality cluster is ACTIVE** (`R1`, `R2`, `RALLY\_DETECTION\_GAP\_CLOSURE`, `RALLY\_QUALITY\_RESEARCH`, `QUALITY\_ITERATIONS`, `REAL\_FOOTAGE\_VALIDATION\_REPORT`, `RALLY\_DETECTION\_QUALITY\_REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)

92 4. **? RESOLVED (2026-07-02) ? finish-archived the hardening stubs.** The three top-level pointer-stubs (`CODE\_AUDIT\_AND\_TEST\_HARDENING`, `HARDENING\_LOOP\_HANDOFF`, `DEFERRED\_HARDENING\_PLAN`) were removed; full archived copies remain under `docs/archives/`, inbound links (README, PLATFORM_ARCHITECTURE, DESKTOP_APP_PLAN) were repointed, and the sweep is summarized in the umbrella's *Prior audits (absorbed)*.

93 5. **`HOSTING\_PLAN.md` / `UI\_PROPOSAL.md` are PROPOSAL but partly overtaken** by ADR-013 (DO dropped) + the shipped khelsutra site. **? `UI\_PROPOSAL.md` banner added (2026-06-21)** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; the `I18N\_PLAN.md` localization track is likewise in implementation there. **`HOSTING\_PLAN.md` banner still pending** (ADR-013 already noted in it).

94 6. **? RESOLVED (2026-06-21) ? `COMPUTE\_DECOUPLED\_SERVING` M0 approved + M1?M4 SHIPPED** (#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 \$100/?10k cap).

95

96 ## 5. Maintenance

97 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].

98

6. user (2026-07-03T16:45:13.345Z)

1 # Architecture Decision Records

2

3 ### ADR-001: Precise shuttle tracking via self-hosted TrackNetV4/WASB on WSL2

4

```

5      **Date:** 2026-05-25
6      **Status:** Accepted
7
8      ### Context
9
10     The pipeline uses an AI multimodal provider (Gemini) for *semantic* rally segmentation.
11     We need **precise frame-level detection of the moving shuttlecock** (and, generically,
12     balls for other sports) to produce higher-quality candidate rallies and a fine-tuning
13     signal.
14
15     We evaluated whether a commercially available model/API could do this. Findings:
16
17     - **Multimodal LLM APIs (Gemini, Claude, GPT-class) are the wrong tool for precise
18       localization.** They reason well about *what* is happening (rally vs. dead time) but
19       cannot reliably track a tiny, fast (~up to 400 km/h), motion-blurred shuttle frame by
20       frame.
21     - The precision leaders are **purpose-built trajectory models**, which are open source
and
22       self-hosted, not big-vendor APIs:
23     - **TrackNetV4** (TensorFlow) ? badminton-specific, heatmap + motion-attention,
24       highest reported shuttle accuracy; ships `src/predict.py` producing per-frame (X, Y)
25       coordinate CSV. https://github.com/AR4152/TrackNetV4
26     - **WASB** (PyTorch) ? multi-sport strong baseline (badminton/tennis/volleyball/?),
27       fits our `SportProfile` abstraction; needs a custom-video inference wrapper.
28       https://github.com/nttcom/WASB-SBDT
29     - True turnkey commercial APIs (Roboflow hosted models) exist but cap below a tuned
30       TrackNet; enterprise systems (Hawk-Eye, SkillCorner, TRACAB) are not API-accessible.
31
32     ### Decision
33
34     1. Adopt **self-hosted TrackNetV4** as the primary precise shuttle detector, with
**WASB**
35       as the multi-sport fallback.
36     2. Run these models under **WSL2 (Ubuntu)**, **not** native Ubuntu. The entire indexer
37       already runs on Windows; WSL2 keeps a single dev environment and quarantines the
38       TensorFlow/PyTorch dependencies. Native Ubuntu would require dual-boot and migrating
39       the whole project for marginal gain.
40     3. Integrate via a **subprocess adapter**: a detector module that shells out to the WSL
41       `predict.py`, reads the coordinate CSV, and converts trajectory ? candidate rallies
42       feeding the existing stitching flow. This keeps the TF dependency fully isolated from
43       the main Python environment.
44
45     ### Consequences
46
47     - **WSL I/O caveat:** `/mnt/c` access is slow. Stage match videos inside the WSL
48       filesystem (e.g. `~/clips/`) before inference rather than reading large mp4s across
the
49       Windows?WSL boundary.
50     - TrackNetV4 (TensorFlow) and WASB (PyTorch) live in **separate conda envs**.
51     - Next step after integration: manual video annotation ? fine-tune the detector.
52
53     ---
54
55     ## ADR-002: Use the sports-data-collector's rally annotations as a golden eval +
temporal-segmenter training set
56
57     **Date:** 2026-05-26
58     **Status:** Accepted
59
60     ### Context
61
62     A sibling project, **sports-data-collector**, curates sports videos and stores per-rally
63     `[start, end)` timestamps (plus stroke counts, scores) in an SQLite DB, with a
64     commercial-license flag per source. We asked whether this data could improve the
65     indexer's rally detection, and for which task.

```

```

66
67     Key findings:
68
69     - The annotations are temporal-only (when rallies start/end). They do not
contain
70     per-frame shuttle `(x, y)` coordinates, so they cannot train the shuttle detector**
71     (TrackNetV4/WASB ? see ADR-001), which needs coordinate labels.
72     - They are an excellent golden evaluation set and a training signal for the
73     temporal rally segmenter** (active vs. dead play over time) ? the stage that is
today
74     a hand-tuned heuristic, not a learned model.
75     - The collector's five sports tables (`badminton_rallies`, `tennis_points`,
76     `cricket_deliveries`, `pickleball_rallies`, `tabletennis_rallies`) all share a
77     `(video_id, ordinal, start_time, end_time)` shape, and the indexer already has
78     `sport_registry` / `segmenter_registry`. So a generic `(start, end)` bridge serves
79     every sport.
80     - The two repos key videos differently: the collector by a human `video_id`, the indexer
81     by file MD5. A re-encode/re-download changes the bytes, hence the MD5.
82
83     ### Decision
84
85     1. Treat the collector's rally timestamps as (a) the golden eval set for rally
86     segmentation and (b) the labels for training a temporal segmenter ? explicitly
87     not as shuttle-detector training data.
88     2. Bridge the repos with a sport-agnostic export (`curate export` in the collector):
89     per-video `start,end` CSVs (the indexer's eval format) + a `manifest.json` that pairs
90     each CSV with its video file. Gate on `is_commercial` so only commercially usable
data
91     flows through.
92     3. Honour the MD5 keying contract: evaluation/processing must run on the exact video
93     file recorded in the manifest; acquisition (ADR-003) therefore precedes processing.
94
95     ### Consequences
96
97     - The collector becomes the producer of labels; the indexer the consumer (processing
+
98     scoring + training). Neither takes a hard dependency on the other beyond the manifest.
99     - Scoring is free and deterministic (reads the DB; no API). See `docs/EVALUATION.md`.
100     - Only invest in frame-level coordinate labels (for the detector) if evaluation
shows
101     detection, not segmentation, is the bottleneck.
102
103     ---
104
105     ## ADR-003: Video acquisition ? always best resolution, upgrade in place, authenticated
cookies
106
107     Date: 2026-05-26
108     Status: Accepted
109
110     ### Context
111
112     The collector's initial crawl downloaded videos with `yt-dlp -f worstvideo` (256x144),
113     mislabelled as 1080p. The indexer needs 720p MP4 for meaningful YOLO/shuttle detection.
114     Re-acquiring full-resolution sources surfaced several hard-won realities:
115
116     - The ShuttleSet corpus tops out at 1080p (no 4K); "best available" resolves to
1080p.
117     - Authenticating with a YouTube Premium account is required for reliable,
unthrottled
118     downloads ? anonymous downloads are throttled (~1.3 MiB/s) and non-deterministic
119     (identical commands variously yielded 1080p DASH, 1080p HLS, or a 144p fallback).
120     - Chrome cookies can't be read on Windows (app-bound encryption / DPAPI, yt-dlp
121     #10927); Firefox cookies read cleanly even while open.
122     - The `web`/`web_safari` player clients are forced into SABR streaming, which strips

```

```

123     download URLs and silently drops to 144p (yt-dlp #12482). yt-dlp's `default` client
124     serves real downloadable 1080p DASH with auth cookies.
125 - Bulk bursts trip YouTube **rate-limiting** (a 22-video burst failed all at once,
though
126     each worked in isolation).
127
128     ### Decision
129
130     1. **Always fetch the best resolution available** (`bv*+ba/b -S res,...`, merged to
MP4);
131     ffprobe the result and store the *actual* width/height/fps ? never assume.
132     2. **Upgrade in place, never clean-slate:** the `fetch` command re-downloads a video and
133     updates only `local_path`/`resolution`/`fps`, preserving every rally annotation and
134     curation status (deleting would cascade-delete the valuable labels).
135     3. Use **authenticated cookies** (`--cookies-from-browser firefox` / `--cookies file`)
and
136     the **`default` player client**; enable the EJS challenge solver (deno).
137     4. **Throttle bulk runs** (`--sleep` + per-video `--retries`/backoff) and treat a
138     below-min-height download as a **failure** (discard, don't commit) so a flaky low-res
139     result never overwrites a good entry.
140
141     ### Consequences
142
143     - All 22 fetchable ShuttleSet matches are now true 1080p (one match, shuffleset_12, has
no
144     source URL in the catalog and remains video-less but keeps its annotations).
145     - Acquisition is reproducible and resumable; the ~26 GB corpus is kept outside git.
146     - Future non-YouTube or 4K sources are supported by the same code (optional
`max_height`).
147
148     ---
149
150     ## ADR-004: Add an offline, API-free *learned* rally segmenter as a first-class mode
151
152     **Date:** 2026-05-26
153     **Status:** Proposed (substrate pending data)
154
155     ### Context
156
157     The default high-quality path (`yolo_hybrid`) refines YOLO candidate windows with
**Gemini**,
158     which costs money and requires network access. With a now-1080p, commercially-licensed,
159     rally-annotated corpus (ADR-002/003), we can learn the refinement step offline instead.
160
161     Crucially, the **API-free signal already exists and is already persisted**:
`yolo_hybrid`
162     Stage 1 is pure YOLO (player tracking ? per-window motion features: peak/mean velocity,
163     active-frame count, track-id count) stored in `candidate_segments.stats`. Gemini is only
164     the Stage-2 refiner. So the rally annotations can **train a classifier to replace
Gemini**,
165     with no external API and **no new labeling**.
166
167     Three candidate substrates were weighed:
168
169     - **Tune `motion` thresholds** on the annotations ? trivial, fully offline, but a low
170     ceiling (pixel motion can't separate rally from crowd/pan/replay). Use as a *floor*.
171     - **YOLO features ? trained classifier** ? reuses already-stored, semantically
meaningful
172     player-motion features; medium-high ceiling; moderate effort. *Recommended primary.*
173     - **WASB shuttle-motion ? rally detector** ? highest ceiling (the shuttle defines play),
174     fully offline but heaviest (needs the WSL2 WASB runner, ADR-001); can't train WASB
175     itself (no xy labels), only a rally layer on its output.
176
177     ### Decision
178

```

```

179     1. Build an offline, API-free learned segmenter as a new registry mode (e.g.
180         `yolo_learned`): label detector windows by overlap with GT rallies, train a
lightweight
181         classifier on the persisted features, emit final segments with zero API calls,
and
182         score it with the existing harness against `motion` and a Gemini reference.
183     2. Choose the substrate from data, not assumption. A learned refiner can only
sharpen
184         the windows the detector found, so the deciding metric is the detector's candidate-
stage
185         recall vs. the annotations: high recall ? build on YOLO features; low recall ?
bring in
186         WASB's shuttle signal to raise the ceiling.
187     3. Keep the Gemini-backed `yolo_hybrid` as an upper-bound reference, not a
production
188         dependency.
189
190     Consequences
191
192     - The final tool gains a self-contained, offline, free rally mode trained on its own
193         curated data ? runnable air-gapped, improvable as more matches are annotated.
194     - The approach is sport-generic: the same label?feature?classifier machinery extends
to
195         tennis/cricket once those have annotations.
196     - The full paid 22-video Gemini evaluation is not run; spend is limited to a single
197         reference video.
198
199     Update (2026-05-26): Phase 3 evidence ? substrate is WASB, not YOLO
200
201     The deciding candidate-recall measurement is in. On shuttlest_8 (64 GT rallies), pure
202     YOLO Stage-1 produced 181 candidate windows with:
203
204     - Recall @ IoU 0.5 = 0.05, but recall @ IoU 0.1 = 0.53 and @ IoU 0.01 = 0.72.
205     - 72% of rallies have some candidate overlap, yet the median best coverage of a
206         rally by any candidate is only ~20%.
207     - Post-hoc merging adjacent candidates does not help (recall stays ~0.05 and
degrades
208         as the merge gap grows ? windows overshoot rather than align).
209
210     Interpretation: YOLO isn't blind to rallies, but its windows are badly misaligned
?
211     because the broadcast camera tracks the players, their on-screen (pixel) velocity
dips
212     during rallies and spikes on pans/cuts/replays. Player-pixel-motion is therefore a
poor
213     rally proxy on broadcast footage (it was designed for static GoPro recordings). A
214     classifier that only filters these windows is capped at ~0.25 recall ? below any
useful
215     bar ? and no simple boundary post-processing recovers it.
216
217     Decision update: per this ADR's data-driven rule (low candidate recall ? bring in
the
218     shuttle signal), the offline segmenter's substrate is WASB shuttle-motion (ADR-001),
219     not YOLO player-motion. The shuttle's motion/presence is camera-invariant and directly
220     defines active play. The `motion` segmenter is likewise unusable here (0 TP ? moving-
camera
221     pixel motion is everywhere).
222
223     Caveats / next: this is one video; the mechanism (camera tracking) generalizes but
224     should be confirmed on 2?3 more via a free, local, candidates-only YOLO pass (no
Gemini).
225     Processing is also slow (~50 min per 1080p video) ? the detection pass likely needs
reduced
226     resolution and/or cached detector outputs to be practical across 22 matches. The
227     substrate-agnostic training-data + classifier layers already built still apply: they

```

```

will
228     consume **WASB-derived** windows/features instead of YOLO's.
229
230     ---
231
232     ## ADR-005: Permissive Dependency Stack for the Hosted Monetization Path
233
234     **Date:** 2026-05-30
235     **Status:** Accepted
236
237     ### Context
238
239     The project will be monetized as a **closed-source hosted API** (users upload videos to
a cloud GPU VM and get highlights). Because we never distribute code/binaries, GPL/LGPL
distribution copyleft is not triggered ? but **AGPL §13** (network clause) and **metered-API
terms** are, and **video-codec patent royalties** are a separate concern from software licenses.
A deep-research audit (2026-05-30, 25 adversarially-verified claims against primary sources ?
see `docs/MONETIZATION_AUDIT.md`) identified the blockers and clean alternatives. This ADR also
**resolves the open licensing question in ADR-001/ADR-004** about WASB.
240
241     ### Decision
242
243     1. **Drop `ultralytics`. Both the library and the `yolov8n.pt` weights are
**AGPL-3.0**; §13 makes them a blocker for a closed-source hosted service. Swapping the codebase
while keeping the `.pt` does **not** escape ? the weights are AGPL too. Replace player/motion-
gating detection with a permissive detector: **torchvision Faster R-CNN / RetinaNet (BSD,
default ? zero new deps, person = COCO class 0)**, escalating to **YOLOX** or **RT-DETRv2 (Baidu
repo, Apache-2.0)** if more speed/accuracy is needed. **Ultralytics Enterprise License** is a
paid fallback only (quote-only pricing).
244     2. **Shuttle tracking stays permissive.** **WASB-SBDT and TrackNetV3 are MIT (CLEAR for
commercial use)** ? this resolves the licensing unknown flagged in ADR-001/ADR-004. **MonoTrack
(Adobe Research, noncommercial) and YOLO-NAS weights (Deci, noncommercial) are banned.** Train
our own checkpoints (or confirm per-checkpoint terms), since distributed research
weights/datasets aren't covered by the MIT *code* grants.
245     3. **LLM step ? keep all three strategies open:** (A) eliminate via an offline temporal
segmenter (**ActionFormer, MIT** ? the Phase-4 direction), (B) self-host **Qwen2.5-VL-7B/32B
(Apache-2.0)** or **InternVL2.5 (MIT)**, or (C) keep **Gemini 2.5 Flash on the paid tier**
(required ? the free tier trains on and human-reviews uploaded video). The `google-genai` SDK
itself is Apache-2.0 (clear).
246     4. **Emit royalty-free output.** Standardize highlight output on **AV1 (+ Opus)** to
avoid AVC/HEVC encode-side patent exposure; provision an **Ada-class GPU (L4/L40)** for hardware
AV1 (NVENC), falling back to **SVT-AV1 (CPU, BSD-3-Clause-Clear)** on T4/A10/A100.
247
248     ### Consequences
249
250     - Eliminates the only two hard legal blockers; the rest of the stack (torch, opencv,
fastapi, numpy, lapx, ffmpeg-as-subprocess, google-genai SDK) is already permissive.
251     - Detector swap is contained to `backend/indexer/yolo_hybrid.py` (the `from ultralytics
import YOLO` load + the `model.track(...)` call) plus `config.json` knobs and
`tests/test_yolo_hybrid.py` mocks. Stage-1 feature outputs (velocity/active-frame/track-id
stats) and Stages 2?4 are detector-agnostic and unchanged.
252     - We now own the frame loop + tracking (ultralytics did both internally). Use a
permissive ByteTrack (e.g. `supervision`, MIT, or the MIT ByteTrack repo) on top of `lapx`
(already a BSD dep).
253     - Residual items needing legal sign-off: exact codec-pool royalty thresholds (AVC/HEVC
decode of uploads), Ultralytics Enterprise pricing, and per-checkpoint weight licenses ? tracked
in `docs/MONETIZATION_AUDIT.md` §7.
254     - The `yolo_hybrid` registry key is now a slight misnomer (no YOLO); keep it for back-
compat or rename in a follow-up.
255
256     ---
257
258     ## ADR-006: Golden-Set Generation via Vendor Bake-off + a Pluggable Annotation Provider
259
260     **Date:** 2026-05-30

```

261 ****Status:**** Accepted (process v1; vendor shortlist pending research)

262

263 **### Context**

264

265 The product needs ****golden-set ground truth**** (hand-verified rally `[start,end)` boundaries) to (a) measure pipeline quality and (b) run ****on-demand / surprise regression tests**** against the production tool. Two forces shape the design: the hosted-API monetization goal raises a "who views the user's video" concern (ADR-005 / MONETIZATION_AUDIT.md), and the founder is ****Bangalore-based**** and prefers ****local/India vendors****. We already own a deterministic interval scorer (`backend/eval/metrics.py`) and a growing collection of ****licensed/owned**** match videos.

266

267 Decisions locked with the owner: ****licensed/owned video only**** (no user uploads to vendors) . ****rally `[start,end)` annotation depth**** . ****tiered turnaround**** (fast automated + slow human) . ****vendor bake-off**** to choose . ****Bangalore/India-local preference****.

268

269 **### Decision**

270

271 1. ****One scorer, three jobs.**** A vendor's `[start,end)` output is structurally identical to a prediction, so `metrics.py` grades product-vs-truth, vendor-vs-key, and annotator-vs-annotator (IAA) with no new scoring code.

272 2. ****Select vendors by measured quality-per-dollar****, via a bake-off: write a precise rally-definition guideline, build a small internal gold key (5?10 licensed clips), send the same clips to 2?3 vendors (favoring Bangalore/India), and grade every submission with our own harness. The numbers are the decision.

273 3. ****Pluggable `AnnotationProvider` registry**** mirroring the existing registry pattern: `FileProvider` (today's manual/ShuttleSet flow), `HumanVendorProvider` (slow tier ? authoritative truth), `AutomatedProvider` (fast tier ? independent model oracle).

274 4. ****Regression = existing harness + one new component.**** `regress(video)` calls a provider for ground truth, runs the production pipeline, and scores with the unchanged `metrics.py` against a per-video `baseline.json`.

275 5. ****Privacy by construction.**** Golden sets use only licensed/owned footage; all third-party video egress flows through the single provider boundary where DPA / no-reuse / delete-on-delivery / access-logging controls live.

276

277 **### Consequences**

278

279 - Eliminates the "human views user video" concern for golden-set/regression work, because user uploads never enter the pipeline.

280 - ****The oracle problem is a first-class caveat:**** the fast-tier `AutomatedProvider` must be a *different model lineage* than production (else shared blind spots make the test falsely green); fast-tier alarms escalate to the slow human tier before blocking a release.

281 - New docs: `docs/GOLDEN\_SET\_VENDORING.md` (the full process v1) and `docs/VIDEO\_RECORDING\_GUIDELINES.md` (a tracked draft: capture-quality thresholds to be derived empirically from precision/recall, since input quality bounds output quality ? cf. ADR-004's broadcast-camera finding).

282 - The vendor shortlist itself is ****pending a primary-source research pass**** (favoring India/Bangalore) and is intentionally left blank until verified ? no un-sourced vendor claims.

283 - ****Honest limitation:**** a licensed/owned corpus may not fully represent real user capture, so regression greens are strong evidence, not guarantees ? the recording-guidelines work aims to narrow that gap.

284

285 **### Update (2026-06-02): shortlist researched; Roora selected for the initial pilot**

286

287 - The ****vendor shortlist**** is filled in (GOLDEN_SET_VENDORING.md §6, primary-source pass 2026-05-30).

288 - The founder selected ****Roora ML Pvt Ltd**** (HSR Layout, Bengaluru) for the ****initial paid pilot****, diverging from the doc's iMerit/Taskmonk default toward a Bangalore-local, startup-friendly shop. Roora's existence, image/video/temporal capability, and location are verified (rooragr.com); ****pilot quality is still to be measured**** ? run as a ****single-vendor pilot first**** (the pilot *is* the calibration), not a multi-vendor bake-off.

289 - ****Step 0 ? the annotation guideline ? is authored**** (the highest-leverage artifact): collector repo `docs/annotation/` ? `R00RA\_BRIEF.md` (rally CSV schema), `ANNOTATION\_GUIDE.md` (illustrated guide), plus intake-manifest and NDA templates.

290 - ****Pilot scope is 3 sports**** (badminton, pickleball, table tennis); the collector's


```

`import-local` ingestion for all three is implemented (collector ADR-001).
291 - **Still pending:** the internal **gold key** (5?10 expert-labeled clips) and the
vendor-dependent **GS-3 `HumanVendorProvider`** ? GS-1/GS-2/GS-4 (FileProvider, regression
runner, automated-oracle scaffold) have already landed (see GOLDEN_SET_IMPLEMENTATION.md).
292
293 ---
294
295 ## ADR-007: Audio "thwack" hit-detection (fused with WASB) is the next rally-detection
cue; pose/stance parked
296
297 **Date:** 2026-06-07
298 **Status:** Accepted (direction); experiments gated on golden eval labels
299
300 ### Context
301
302 After building the WASB shuttle-trajectory substrate (ADR-001), a Gemini-teacher ?
local-student distillation
303 (ADR-004 lineage), and measuring both, two honest findings emerged: **(a)** threshold-
tuning the local
304 windowing ceilings around F1 ? 0.44 and the learned student *underperforms* the
threshold baseline at n=2
305 labeled videos; **(b)** the dominant failure is **candidate RECALL** ? WASB-derived
windows simply don't align
306 with (or miss) true rallies (IoU-0.5 recall ceiling ~0.43 on one clip). To decide
whether teacher?student was
307 even the right backbone, we ran a deep-research survey
(`docs/RALLY_DETECTION_RESEARCH.md` ? 23 sources, 25
308 adversarially-verified claims). Verdict: **teacher?student is sound but NOT uniquely
best**; the strongest
309 direction is a **complementary, fully-local multi-cue stack**, and the **#1 cheapest
lever is audio hit
310 detection fused with the trajectory** (audio+visual fusion measurably lifts rally
*recall* ? tennis HMM 83%
311 fused vs 54% visual / 63% audio). Pose/court serve-gating is the validated #2 but is
engineering-heavier.
312
313 ### Decision
314
315 1. **Adopt a fully-local AUDIO hit-detection cue** (shuttle "thwack" impacts) and **fuse
it with the WASB
316 trajectory** ? primarily to recover rallies WASB misses (recall) and secondarily as
features for the
317 distilled student. Detailed design: `docs/AUDIO_RALLY_DETECTION_PLAN.md`.
318 2. **Gate on an empirical GO/NO-GO probe (E1).** Shuttle impacts are quiet vs tennis/TT,
so before building
319 anything heavy, measure on *real GoPro audio* whether hit-clusters align with golden
rallies and recover
320 WASB misses. Build E2/E3 only if E1 passes.
321 3. **Keep Gemini as an OFFLINE teacher only** (never runtime) ? research confirms this
is the correct role for
322 a cloud VLM; it is not the deployed localizer.
323 4. **Park pose/court-geometry stance detection as the #2 cue** (serve-ready START +
flight-gradient END). It is
324 expected to be substantially harder to get right; revisit after audio (and a TBD
owner hunch).
325 5. **Make "audio always enabled" a capture requirement**
(`docs/VIDEO_RECORDING_GUIDELINES.md`), since the cue
326 is only as good as the recording.
327 6. Stay permissive (ADR-005): classical DSP (numpy + stdlib `wave`, BSD), no restrictive
weights; any later model
328 trained in-house.
329
330 ### Consequences
331
332 - New `backend/pipeline/audio/` module (extraction + onset/peak + confidence), feeding

```

```

`distill_local` and scored
333     by the existing `metrics.py` / golden loaders. *(As built: the detector is **pure
numpy + stdlib `wave`, no
334     scipy** ? the anticipated scipy dependency proved unnecessary.)*
335     - **All audio measurement is blocked on golden eval labels** ? which the VLC rally-
annotator (now a standalone
336     MIT, multi-sport open-source repo: https://github.com/avidullu/rally-annotator) lets
the owner self-produce,
337     and the in-repo `--labels` loaders / collector `import-local` ingest.
338     - Sport-generic: tennis/table-tennis hits are *louder*, so the cue should transfer at
least as well there.
339     - **Honest risk:** if E1 shows shuttle SNR is too poor on real amateur audio, audio is
deferred ? this ADR
340     commits to the *experiment and direction*, not to a guaranteed win.
341
342     ### Integration architecture: modular fusion now (Option A), an owned end-to-end model
later (Option B)
343
344     A recurring design question: do we **feed audio into WASB** so it refines its own
output, or **keep audio as a
345     separate signal fused in our code**? The answer today is the latter ? and understanding
*why* sets up the
346     long-term route.
347
348     **Why fusion lives in our rally layer, not inside WASB (Option A ? now).** WASB is a
**frozen, vision-only
349     shuttle *detector*** (image frames ? per-frame shuttle `(x, y, visibility)`; MIT weights
we *run*, not train).
350     It has no audio pathway and it does not decide rallies at all ? **rally detection is our
own layer** on top of
351     its trajectory (`trajectory_to_action_windows` ? gate ? the distilled classifier).
Making WASB consume audio
352     would mean bolting on an audio encoder and **retraining the network**, which needs per-
frame
353     shuttle-coordinate-*plus*-audio labels we don't have (we only have rally `[start,end]`
labels) and would fork
354     the clean MIT model. It also conflates two different questions: WASB answers ***where*
the shuttle is**; audio
355     answers ***when* a hit happened**. Audio doesn't make WASB localize better ? it's an
independent rally-activity
356     cue. So fusion belongs where cues are *combined into rallies*: our layer.
357
358     ``
359     frames ?? WASB (frozen, MIT) ?? shuttle trajectory ??
360     audio ?? hit detector (ours) ?? hit events ?????????? RALLY LAYER (ours) ?? rallies
361     [pose ?? stance heuristic (ours; the parked #2) ???? (feature + decision fusion)]
362     ``
363
364     Two complementary fusion modes (both in `docs/AUDIO_RALLY_DETECTION_PLAN.md`):
365     1. **Feature-level (primary):** audio features (hit_rate, inter-hit cadence/regularity,
time-since-last-hit)
366     join WASB-derived features (velocity, net-crossings) in the **single distilled
classifier** that emits
367     rallies ? "use audio to sharpen our rally metrics" (experiment E3).
368     2. **Decision-level recall-recovery:** where WASB proposed nothing but a dense hit-
cluster exists, propose a
369     rally there ? directly correcting WASB's recall misses.
370
371     Each cue (WASB, later the person/pose detector) stays a **swappable black box** ; every
signal is independently
372     testable ? which matters while we're label-starved.
373
374     **The long-term route (Option B ? the intended destination, once a healthy golden corpus
exists).** As we
375     accumulate enough golden eval/training videos (owner self-rating via the annotator +

```

vendor labels), the right

376 end-state is to ****collapse the modular cues into a single OWNED multimodal rally**

model**, trained/fine-tuned

377 end-to-end on (frames + audio + pose/stance + ?) ? rally boundaries ? the HitNet /

temporal-action-localization

378 direction surfaced in `docs/RALLY\_DETECTION\_RESEARCH.md`. Crucially, that lets us

****fine-tune from our own code**

379 and data instead of depending on WASB's pretrained weights** ? removing both the ***recall**

ceiling* imposed by

380 WASB's candidate windows and the ***licensing question*** on its academic-trained weights.

381

382 ****Why we don't jump to Option B now ? and why Option A is the path to it.**** Option B

needs data we don't yet

383 have, and it's far harder to debug as one black box. So we deliberately build ****modular**

cues first** ? audio

384 (this ADR), then a stance heuristic (the parked #2), then other signals ? each cheap,

debuggable, label-

385 efficient, and ***each one generating the very golden labels an end-to-end model will**

need*. When the corpus is

386 healthy, we train the unified model. ****Option A is not a detour from Option B; it is how**

we earn it.**

387

388 **## ADR-008: Owned-model topology ? artifact boundary now, repo split at Gen-2**

389

390 ****Date:** 2026-06-10**

391 ****Status:** Accepted (owner-ratified after an adversarial design panel)**

392

393 **### Context**

394

395 M0.4 (Gen-0 TAL train?eval?ship-gate harness, productionizing

`backend/eval/rally\_seq\_proto.py`

396 over ActionFormer/ASFormer) and M0.1 (corpus spine + event-index contract) are greenlit.

The

397 question: should the model/training work live ****in this repo****, or in a ****separate**

repo** whose

398 output the indexer consumes as another segmenter (like WASB and Gemini)? The owner's

instinct

399 was isolation ("iterate on model and processing independently"), citing the collector

split as

400 precedent.

401

402 **### Decision**

403

404 ****Draw the boundary at the ARTIFACT CONTRACT now; split the repo at Gen-2, as a pre-**

committed,

405 trigger-based event ? not today.**

406

407 1. ****Training lives in this repo as a top-level `training/` tree**** (sibling of

`backend/`),

408 isolated ***executably***, not socially: its own `training/requirements-train.txt`

(heavy/

409 training-only deps never enter `requirements.txt`), and the import-smoke suite

asserts

410 ``import backend.main` loads **no `training.` module**.`

411 2. ****The featurizer is single-source-of-truth****: promoted out of `rally\_seq\_proto` into

412 ``backend/features/rally_seq.py`` (pure numpy/scipy, versioned schema constant),

imported by

413 BOTH the training harness and the future TAL serving segmenter. Train == serve by

414 construction ? the fps-normalization bug (fixed 2026-06-10) is the skew class this

kills.

415 3. ****Serving consumes only a versioned ARTIFACT BUNDLE**** (weights gitignored / private

GitHub

416 Releases + a manifest carrying: feature-schema version, normalization, calibration/

417 post-processing, training-corpus ref + GT hash, consent + license attestation incl.

the

```

418 WASB C3 checkpoint status). The loader/gate hard-fails on schema mismatch ? the
419 `regression.py` incommensurability-guard pattern.
420 4. **The ship-gate is owned by the product side** (extends `backend/eval/regression.py`
+
421 `metrics.match_intervals`): the trainer runs it as pre-flight but never grades
itself.
422 The gate refuses commercial ship while C3 is unresolved, regardless of F1.
423 5. **Pre-committed split trigger** ? extract `training/` into a separate repo (working
name
424 `sports-temporal-models`, to be finalized with the pending repo rename) **when**:
425 (a) Gen-2 arrives (ms-swift / Qwen3-VL / cloud-GPU training ? an alien dependency
tree), OR
426 (b) a second consumer of the trained artifacts appears (e.g. the khelacharya layer),
427 AND (c) the harness's only data input is the collector's M0.1 export. By then the
artifact
428 contract is proven and extraction is a file move across a known seam.
429
430 **Prerequisite frozen first: the golden-label/corpus contract.** One canonical GT loader
(the
431 repo carried two incompatible ones: `annotations.load_annotations` `start,end` vs
432 `calibrate_wasb.load_golden_gts` `start_time,end_time`, the latter feeding 7 modules) +
a
433 round-trip test against the collector's `curate export` format. Every ship-gate quantity
434 (gt_hash, LOVO numbers, the future consent fingerprint) flows through this boundary.
435
436 ### Rationale (the decisive, code-verified facts)
437
438 - **The "just another runner like WASB/Gemini" analogy fails at the seam location.**
WASB's
439 contract sits at the video boundary (video ? trajectory CSV); Gemini's at video ?
segments.
440 The Gen-0 head consumes **features this repo computes** from the trajectory ? the seam
is
441 mid-pipeline. Since the indexer is an app (not pip-installable), a separate repo today
could
442 share that code only by **copying** it ? this project's one empirically observed
failure
443 mode (the GT-header fork; the duplicated TUNED constants).
444 - **The in-house separate-repo precedent is unproven for load-bearing deps**: the
`obsreport`
445 pin has never been live (commented in requirements.txt, repo unpublished, CI skips via
446 importorskip). It works because its absence is a harmless no-op ? a property a
featurizer
447 can never have.
448 - **Gen-0 has no environmental forcing function** (trains in minutes, ~4.5 GB, MIT pip-
able
449 deps, on the owner's WSL/2070S box); Gen-2 genuinely does. Splitting then is a
scheduled
450 event, not an overruled instinct.
451 - The prerequisite work for a clean split (featurizer promotion, loader consolidation,
452 manifest) is **identical** to the in-repo design ? doing it in place first makes the
453 eventual extraction cheap and safe. This is "envision now, build later" applied to
topology:
454 **the split is driven by productionization needs, not code-isolation aesthetics**
(owner's
455 framing, ratified).
456
457 ### Consequences
458
459 - `backend/` may never import `training.*` (CI-enforced). Training tests use **synthetic
460 fixtures only** until the M0.1 consent ledger exists (no real trajectories/labels
committed).
461 - Weights are never committed; artifact manifests are the reviewable/committable
surface.
462 - The collector remains system-of-record for labels + consent; the indexer owns scoring

```

```

and
463     the ship/no-ship decision; manifests pin both by hash.
464 - Revisit triggers are explicit (above); if agent-driven development repeatedly breaches
the
465     import wall despite CI, that is evidence the social+test boundary failed and the split
466     should happen early.
467
468     Panel artifacts: workflow `wf_471a9cdd-d1d` (3 grounded position papers + cross-
examination),
469     2026-06-10.
470
471     ## ADR-009: KhelSutra Guru ? product identity, web-UI repo topology, local-first
delivery
472
473     **Date:** 2026-06-10 . **Status:** RATIFIED (owner, in-session)
474
475     ### Decision
476
477     1. **Product name: KhelSutra Guru**; domain **khelsutra.guru** (GoDaddy receipt: 3-yr
478         registration + Full Domain Protection, renews 2029-06). Earlier "khelstura" spellings
were typos.
479     2. **The web product lives in a separate private repo `khelsutra`** ("the heart and
brain"):
480         `web/` SPA (alpha), `control-plane/` at beta, product docs migrating in over time.
The backend
481         enablers (async jobs #16, chunked upload/download, identity scoping, correction
endpoints)
482         stay in the indexer repo ? they are engine API features.
483     3. **Local-first delivery:** every alpha feature is built and end-to-end-tested against
484         localhost before any cloud/DNS/deployment work starts (gates GL ? GD in
485         `docs/UI_ALPHA_EXECUTION_PLAN.md`).
486     4. Alpha topology per `docs/HOSTING_PLAN.md`: free edge (Cloudflare Pages/Access/Tunnel)
+ the
487         owner's GPU box; GCP credits reserved for the beta control plane (Cloud Run/GCS).
488
489     ### Rationale
490
491     - Repo split: the npm/UI dependency tree stays outside the indexer's MIT/Apache/BSD
licensing
492     gate; Pages auto-deploy wants a repo boundary; the engine repo's pending rename stops
blocking
493     product work. Named `khelsutra` (not `khelsutra-ui`) so the beta control plane doesn't
force a
494     third repo.
495     - Local-first (owner-requested): removes cloud variables while the new job/upload
surface
496     stabilizes; deployment becomes a configuration exercise against a proven build.
497     - Planning docs stay canonical in the engine repo for now (single decision ledger; the
copy-drift
498     failure mode is documented in ADR-008's rationale) with pointer READMEs in
`khelsutra`.
499
500     ### Consequences
501
502     - The UI alpha greenlights DEFERRED_HARDENING_PLAN **#16** as PR-1; the owner still
gives an
503     explicit "go" before coding starts (CLAUDE.md owner-gating).
504     - `khelsutra` is all-rights-reserved (no OSS license file); secrets never committed;
Pages build
505     root = `web/`.
506     - Alpha auth = Cloudflare Access JWT at the edge; the backend reads the authenticated
email;
507     local dev uses a `DEV_USER_EMAIL` override ? no auth code lives in the SPA.
508     - Sandbox note: git metadata operations on the Cowork mount are unreliable (lock/unlink
denied);

```

```

509     `khelsutra` git init/commit/push run Windows-side (owner or Claude Code).
510
511     ## ADR-010: DECOUPLED_COMPUTE ? pivot GPU + storage from DigitalOcean to GCP (asia-
512     south1)
513     **Date:** 2026-06-20 . **Status:** RATIFIED (owner, in-session)
514
515     ### Decision
516
517     1. The DECOUPLED_COMPUTE **GPU box + co-located object storage move from DigitalOcean to
518     GCP**,
519     region **asia-south1 (Mumbai)** (owner picked Mumbai over asia-southeast1/Singapore ?
520     closest GCP
521     region to Bangalore ? lowest SSH/upload latency). GPU + bucket in ONE region = no
522     cross-DC reads.
523     2. **Storage = GCS** bucket `gs://khelsutra-rally-corpus` (asia-south1); **compute = a
524     GCP GPU VM**
525     (T4/L4) in the same region. The `gcs://` storage backend (already built) is the
526     storage layer.
527     3. **Project = existing `gen-lang-client-0306026826`** (billing **"Khelsutra Billing"**)
528     (`01420D-419EE0-53D83C`). Provisioning runs via gcloud; every resource + a
529     teardown/switch-off
530     path is documented in `deploy/gcp/` (`README.md` + `provision.sh` + `teardown.sh`).
531
532     ### Rationale
533
534     - **DigitalOcean GPU Droplets are North-America-only** (`nyc2`/`tor1`/`atl1`) ? DO has
535     **no
536     Singapore/India GPU** (`SETUP_NEW_MACHINE.md` §2 lists this). The prior session's
537     CPU/stub e2e ran
538     droplet-in-`blr1` + Spaces-in-`sgp1` ? cross-DC latency + egress. GCP has GPUs in
539     asia-south1
540     (Mumbai) + asia-southeast1 (Singapore), enabling true GPU+bucket co-location in Asia.
541     - Existing project chosen over a fresh one to avoid billing-link friction (billing
542     already attached).
543
544     ### Consequences
545
546     - The old **DO CPU droplet (id `579001481`, `blr1`) is drained** (`/srv/khelsutra-site`
547     copied off)
548     **+ destroyed** ; the **exposed DO Spaces key is rotated** (it leaked in chat).
549     - **New-project GPU gate:** `GPUS_ALL_REGIONS` quota = 0 must be raised to ?1 (owner
550     console
551     request); regional asia-south1 `NVIDIA_T4_GPUS`/`NVIDIA_L4_GPUS` are already 1. A bare
552     Gemini/AI-Studio project also needs a one-time console enable of the **Service Usage
553     API** before
554     any gcloud provisioning works (gcloud's own `services enable` needs it ? chicken-and-
555     egg).
556     - **Cost guardrails:** a `$100` budget alert (`rally-100usd-guard`, 50/90/100%),
557     cheapest adequate
558     GPU (**T4**), and STOP/DELETE the VM when idle (`deploy/gcp/teardown.sh`). The bucket
559     is never
560     auto-deleted.
561     - `SETUP_NEW_MACHINE.md` §2's "DO GPU droplet in `sgp1`" is **superseded** by the GCP
562     path
563     (`deploy/gcp/README.md`). **WASB-C3** (weight provenance) remains the open legal item
564     before
565     sharing any trajectories/model externally.
566     - The M3.5 **native WASB runner** that makes the GPU box useful landed in **PR #256**
567     (`detector_impl="native"); its byte-parity acceptance gate is verified on the GPU
568     box, not in CI.
569
570     ## ADR-011: DECOUPLED_COMPUTE scope = decoupling core (M0?M2 + M3.5); defer
571     serving/billing; DO=training, GCP=serving
572

```

553 **Date:** 2026-06-20 . **Status:** RATIFIED (owner, in-session). **Refines ADR-010.** ?
 554 **The DO=training / GCP=serving split is SUPERSEDED by [ADR-013](#adr-013-drop-digitalocean-for-compute--gcp-is-the-sole-compute-stack-training--serving)** ? DigitalOcean is dropped for
 555 compute; GCP is the sole compute stack. (The scope decision ? DECOUPLED = M0?M2 + M3.5 ? still
 556 stands.)

557 ### Decision

558 1. **DECOUPLED_COMPUTE's definition-of-done = the decoupling CORE: M0 (CPU e2e) + M1
 559 (single GPU
 560 run) + M2 (train-from-bucket) + M3.5 (Linux-native detector parity)** ? i.e. "the
 561 rally pipeline
 562 runs off the local box on a rented GPU, bucket-driven train + infer, with real
 563 detectors." It is
 564 **archivable after ONE real GPU run** proves M1/M2/M3.5 on cloud hardware.

565 2. **DEFER M3 (inference serving endpoint + per-video billing) and M4 (scale-to-zero +
 566 cost-guard)**
 567 out of DECOUPLED_COMPUTE ? into the **monetization / appliance (khelsutra) track.**
 568 They are
 569 productization (a cost ledger on the `jobs` table + a non-owner-upload DPA/consent
 570 gate), not
 571 compute-portability.

572 3. **Stack split by purpose** (refines ADR-010, which had moved *everything* to GCP):
 573 - **DigitalOcean = TRAINING / the decoupling core.** Use the **\$200 DO credits** ; DO
 574 GPU droplets
 575 are North-America-only (`tor1`/`nyc2`) + co-located DO Spaces (`nyc3`) ? NA co-
 576 location is fine
 577 for training/dev (no per-customer latency requirement). This is where M1/M2/M3.5
 578 get proven.
 579 - **GCP = the SERVING path to real customers.** Cloud Run GPU (scale-to-zero) gives
 580 native
 581 per-invocation GPU-seconds = the exact per-customer/per-video bill (M3/M4,
 582 monetization track).
 583 The GCS bucket (`gs://khelsutra-rally-corpus`) + SA (`rally-worker`) + `\$100`
 584 budget provisioned
 585 under ADR-010 are **RETAINED** for this serving path (do NOT tear them down).

586 ### Rationale

587 - The North Star question ("can we lift compute onto a rented GPU doing bucket
 588 train+infer?") is
 589 answered by M0?M2 + M3.5; M3/M4 answer a *different* question ("bill a non-owner
 590 customer per
 591 video"), which is serverless-shaped and the docs already route to Cloud Run/RunPod,
 592 not DO.

593 - Splitting by purpose uses each provider's strength: DO's real credits + 1 TiB-egress
 594 Spaces for the
 595 heavy training pulls; GCP's scale-to-zero per-invocation metering for customer
 596 billing. (DO has no
 597 serverless GPU ? coarse wall-clock billing only; unsuitable for per-customer serving.)

598 ### Consequences

599 - **Archive `docs/DECOUPLED\_COMPUTE/` only AFTER the DO GPU run is green** (M1/M2/M3.5)
 600 ? not
 601 aspirationally. Until then it stays a live project doc with this narrowed scope.

602 - M3.5 acceptance is **parity within ~1e-3 px** (not strict byte-equality), validated on
 603 an RTX 2070
 604 (native==WSL 1194/1195; deterministic) ? see PR #259. **Spot does NOT bypass
 605 `GPUS\_ALL\_REGIONS`**
 606 (tested) ? both on-demand + Spot need the quota; a `GPUS\_ALL\_REGIONS?1` increase was
 607 filed via the
 608 Cloud Quotas API (pending). GCP GPU quota matters for the GCP *serving* spike, not the
 609 DO training run.

610 - Small carried-forward code item: de-hardcode `output` in

```

`trajectory_hybrid.py:237,313` (bucket
592     cleanliness). **WASB-C3** (weight provenance) follows the model into the monetization
track (gates
593     *sharing* a trained model, not internal decoupling).
594     - The D0 training run needs: a D0 API token (owner), the rotated Gemini key, WASB
weights, corpus
595     staged to Spaces. The old D0 CPU droplet `579001481` is drained + destroyed; the
exposed Spaces key
596     rotated (ADR-010).
597
598     ## ADR-012: Compute = GPU-native (L4/T4); TPU deferred; Cloud Run + GPU (scale-to-zero)
is the supported serving model
599
600     **Date:** 2026-06-20 . **Status:** RATIFIED (owner, in-session). **Refines ADR-011**
(concretizes the "GCP = serving" path).
601
602     ### Decision
603
604     1. **Stay GPU-native (NVIDIA L4/T4) for all compute** ? local box, GPU VM, and the
serving path ?
605     for the foreseeable roadmap. **Do NOT invest in TPU now.** Revisit TPU **only**
if/when a large
606     *owned* transformer (Qwen3-VL Gen-2+, owner-gated) at *sustained batch* scale becomes
the dominant
607     cost, and **only after a benchmark** proves it ? never speculatively.
608     2. **Cloud Run + GPU (L4), scale-to-zero, is the supported SERVING deployment model**
for the
609     pay-per-video orchestration economics ("users pay us to orchestrate on our GCS bucket
+ VMs /
610     Cloud Run"): pay-per-request GPU-seconds = the exact per-video bill, zero idle-GPU
cost, co-located
611     with `gs://khelsutra-rally-corpus`. This is the concrete form of ADR-011's deferred
**M3/M4**.
612     3. **The named optimization lever is GPU-accelerated video decode (NVDEC)**, not more
GPU FLOPs ? the
613     measured bottleneck is video *decode*, not the model.
614
615     ### Rationale
616
617     - **Workload isn't TPU-shaped.** WASB is a tiny (5.8 MB) PyTorch CNN doing per-frame
heatmap
618     inference; the owned model is a VLM. TPUs win on large, dense, *batched* matmul at
training scale;
619     streaming per-frame CNN inference with small/dynamic batches is a poor fit.
620     - **The real bottleneck is video decode, which TPU does nothing for.** Measured on the
2026-06-20 GCP
621     L4 M2 run: cv2 PNG extraction at ~12 fps with the **GPU idle during extraction**. The
lever that
622     helps is **NVDEC** (hardware decode on L4/T4) + parallel decode ? i.e. lean *into* the
GPU.
623     - **Portability / single-runtime.** TPU needs XLA (torch_xla / JAX) ? significant
fragile NRE to port
624     WASB (torch 1.11) + the pipeline, a second runtime to maintain, and **GCP-only** ? it
would break
625     the local ? VM ? Cloud Run parity the DECOUPLED storage/compute seams exist to
provide. One portable
626     PyTorch+CUDA image that runs identically everywhere is worth more than peak FLOPs
here.
627     - **Economics.** Cloud Run+GPU scale-to-zero meters per request = clean per-video
billing; TPU pricing
628     (v5e pods / queued / committed-use) targets sustained training and its regional
availability is
629     *narrower* than GPU (the M2 run already hit a Mumbai GPU stockout ? landed in
Singapore; TPU would be
630     worse).

```



```

631
632     ### Consequences
633
634     - The **deferred M3 (inference serving endpoint + per-video billing) and M4 (scale-to-
635       guard)** from ADR-011 become a **new tracked subproject ? "Cloud Run + GPU serving"
636       ? to be spun
637       up (as a `docs/PROJECT_DOC_TEMPLATE.md`-based doc) **AFTER `docs/DECOUPLED_COMPUTE/`
638       is archived**
639       (owner sequencing). Seed scope below. **? REALIZED in ADR-014 (2026-06-21):** the
640       subproject is
641       `docs/COMPUTE_DECOUPLED_SERVING/`, **renamed** to reflect that **truly-local is a co-
642       equal profile**
643       (not just cloud serving); ADR-014 expands this 7-point seed into the M0?M10 plan.
644     - **First subproject work item = containerize `python -m backend.worker`** for Cloud
645       Run. This is the
646       known hard part: WASB's torch-1.11 conda env makes the image non-trivial (two runtimes
647       ? the
648       indexer venv + the `wasb` conda env). The decoupled storage/compute seams + the native
649       runner
650       (`detector_impl="native"`, PR #256) make the *code* portable; packaging is the gap.
651     - Keep the **storage/compute decoupling seams** intact so the SAME container runs local
652       / GPU VM /
653       Cloud Run (the per-video workspace convention, in progress, reinforces this).
654     - **Edge auth is a hard prerequisite** before any public serving (engine API is CORS
655       `*`, no auth) ?
656       carried from the appliance track.
657     - **Region/capacity reality:** GPU stockouts are common (Mumbai was fully stocked-out on
658       2026-06-20);
659       scale-to-zero + multi-region/flexible-zone selection is part of why Cloud Run+GPU
660       fits.
661     - TPU is explicitly **out of scope**; this ADR records *why* so it isn't relitigated.
662
663     ##### Seed scope for the "Cloud Run + GPU serving" subproject (formalize post-archival)
664     1. Containerize `backend.worker` (indexer venv + `wasb` conda env; NVDEC-capable base
665       image).
666     2. Cloud Run + GPU (L4) service: `infer` over a `gcs://` workspace; scale-to-zero.
667     3. Per-video cost ledger (on the `jobs` table) = the per-customer bill (M3).
668     4. Cost guard + concurrency caps (M4).
669     5. Edge auth (tenancy boundary) before public exposure.
670     6. NVDEC GPU-decode to kill the extraction bottleneck.
671     7. DPA / consent gate for non-owner uploads (productization, not compute).
672
673     ## ADR-013: Drop DigitalOcean for compute ? GCP is the SOLE compute stack (training +
674     serving)
675
676     **Date:** 2026-06-20 . **Status:** RATIFIED (owner, in-session: "drop DO and document
677     it? record it and bake it for all future work"). **Supersedes the DO=training split in ADR-011;
678     refines ADR-010 + ADR-012.**
679
680     ### Decision
681
682     1. **DigitalOcean is DROPPED as a compute provider.** **GCP is the sole compute stack**
683       for BOTH training and serving. (ADR-011's "DO=training, GCP=serving" split is retired ? GCP does
684       both. The DECOUPLED scope decision in ADR-011 ? M0?M2 + M3.5 ? still stands.)
685     2. **Standing default for ALL future compute work = GCP** (a GPU VM today via
686       `deploy/gcp/`; Cloud Run+GPU for serving per ADR-012). Do **not** re-evaluate DO/Paperspace for
687       compute without a new ADR.
688     3. The `$200` DO credits, if used at all, go **only to non-compute** that DO bills
689       cleanly (CPU droplets / a site host / Spaces storage / bandwidth) ? **never GPU**.
690
691     ### Rationale (researched 2026-06-20, against DO's own docs)
692
693     - **DO GPU is neither enabled on the account nor a clean free-credits path.** The DO API
694       `v2/sizes` returns **0 GPU sizes** (GPU Droplets not enabled); GPU Droplets are **North-

```

America-only** (NYC2/TOR1/ATL1/RIC1/AMS3 ? no India/Singapore); and D0's own docs state that for **newer customers D0 charges the team's primary payment method** when GPU Droplets are created (anti-abuse) ? so GPU spend hits the **card**, not the promo credit. The ADR-011 premise ("200 credits fund D0 training") does **not** hold.

674 - **Paperspace (now D0 Gradient) free GPU is subscription-gated** (a paid "G"/Growth plan; high-end A100/H100 needs the \$39/mo Growth plan), and Paperspace bills **separately** from D0 Droplet credits ? the \$200 credit doesn't fund it either.

675 - **GCP works today:** quota granted (`GPUS\_ALL\_REGIONS=1`), bucket + SA + `\$100` budget provisioned, and M1/M2/M3.5 ran on an L4 for ~\$12/run with no subscription. Real money, but trivial per run ? and **one** portable PyTorch+CUDA stack (local ? GPU VM ? Cloud Run), no second provider to maintain.

676

677 ### Consequences

678

679 - **Compute provisioning = `deploy/gcp/` only.** `SETUP\_NEW\_MACHINE.md` repointed to GCP; `deploy/digitalocean/README.md` marked **DEPRECATED**; `HOSTING\_PLAN.md`'s "All-DigitalOcean" compute option retired.

680 - **`deploy/digitalocean/` scripts are KEPT** ? `bootstrap.sh` / `mount\_scratch.sh` / `gpu\_setup.sh` are **provider-agnostic** Linux-GPU setup the GCP runbook reuses; only the D0-specific *provisioning* (doctl droplet create, Spaces) is dropped. (Optional later cleanup: move them to `deploy/common/`.)

681 - The shared D0 CPU droplet **`claude-do-rally-test`** (another session's testbed) is **owner-coordinated** cleanup, separate from this decision; the freshly-rotated **D0 API token** can be retired once that droplet is gone.

682 - **Storage stays bucket-abstracted** (`gcs://` primary). The `s3` backend still speaks any S3-compatible store, so **D0 Spaces remains a *possible* storage target** even though D0 *compute* is dropped.

683 - **Capacity reality:** GCP GPU stockouts are common (asia-south1/Mumbai was fully stocked-out 2026-06-20 ? the M2 run landed an L4 in asia-southeast1/Singapore; weights still land in the asia-south1 bucket). Sweep zones/regions; scale-to-zero (ADR-012) is part of why Cloud Run+GPU fits.

684 - **DECOUPLED_COMPUTE** archives on M2 completion with a note that the D0 leg was **dropped** (this ADR), not "delivered on D0".

685

686 ## ADR-014: Compute-decoupled rally serving ? one pure core, profile-selected compute + storage

687

688 **Date:** 2026-06-21 . **Status:** ? **RATIFIED** (owner, 2026-06-21 ? M0 design sign-off; build M1+). **Refines ADR-011 + ADR-012** (it *realizes* their deferred M3/M4; it does **not** supersede them); **builds on ADR-013** (GCP sole compute stack). Full design: `docs/COMPUTE\_DECOUPLED\_SERVING/`.

689

690 ### Evolution / lineage (trace the idea)

691 `ADR-005` (permissive licensing ? engine + image stay MIT/Apache/BSD) . `ADR-008` (train==serve featurizer single-sourcing ? the parity discipline this extends to local?cloud) . `ADR-009` (**local-first delivery** ? origin of the truly-local profile) . `ADR-010` (DECOUPLED D0?GCP) . `ADR-011` (**deferred M3 serving + M4 billing**; overrode "rent nothing / not a service"; \$06 gates = DPA/consent, collector `md5`, WASB-C3) . `ADR-012` (**Cloud Run+GPU = serving model**; NVDEC lever) . `ADR-013` (**GCP sole compute stack**) . **ADR-014** (this). Non-ADR: realizes `PLATFORM\_ARCHITECTURE.md`'s `ComputeBackend`/`StorageBackend` registries; the truly-local profile = `VIDEO\_LOCALITY\_MODEL.md` L0 + `DESKTOP\_APP\_PLAN.md` `LocalDesktop`; informed by the archived `DEPLOY\_REPORT\_GCP\_2026-06-20` (cross-GPU sub-pixel parity finding).

692

693 ### Decision

694

695 1. The rally engine is a **PURE function of (input bytes + config) ? outputs**, in three deterministic stages ? **DETECT** (`DetectorRunner.run\_predict`?CSV), **WINDOW** (`trajectory\_to\_action\_windows`, CPU-pure), **STITCH** (`VideoCompiler.compile\_highlights`, ffmpeg) ? that are **NEVER branched on the deployment profile**.

696 2. A **`ComputeBackend` / deployment profile** decides **WHERE** the core runs (local process vs a Cloud Run GPU job); a **`StorageBackend`** decides **WHERE I/O comes from** (local FS / USB / mounted Google Drive / bucket). **Both are selected by ONE startup/setup config** (`deployment.profile` + `compute\_target`), realizing the PLATFORM_ARCHITECTURE `local|hosted|byo\_worker` registry.

697 3. Ship at least two profiles: **truly-local** (local GPU + **in-process stitch**, no

cloud) and **cloud-serving** (a Cloud Run GPU job runs detect **AND** stitch), plus **cpu-mock** for CI and a reserved, **deferred** `byo_worker` hybrid.

698 4. In cloud-serving, **STITCH** runs on Cloud Run (co-located with detect) so its CPU/wall-time + egress are **metered** into a per-job cost ledger alongside `gpu_active_seconds` ? stitching is compute-intensive and its cost must be accounted for. (Truly-local stitch is intentionally **unmetered**.)

699 5. **Parity** is enforced at **WINDOW + STITCH** (byte-exact for a fixed trajectory); **DETECT** is byte-exact only **same-GPU** ? cross-GPU is sub-pixel-different (deploy report 2026-06-20), so it is judged at the **rally/window level**, never CSV bytes.

700 6. Land **smallest-safe-first**, one PR per milestone (M0?M10), **DEFAULT-OFF** where a change touches core callers (e.g. the configurable output base), each guarded by golden-fixtures + a **profile-parity test**; default flips ship a **Launch Report**.

701

702 **### Rationale**

703

704 - The seams already exist: `_resolve_runner` is the single compute seam, the `StorageBackend` ABC + 4 backends + the worker fetch/put are built, and the stub detector (`replay_csv`) gives a **\$0 deterministic** core. Realizing a profile system is **composition**, not a rewrite.

705 - Separating **WHERE** (ComputeBackend, coarse, per-job) from **HOW** (DeviceContext, fine, per-GPU) from **WHAT-bytes** (StorageBackend) keeps each axis independently testable and avoids the main trap ? conflating them.

706 - **Metering stitch on Cloud Run** is the only way the owner's real serving cost (GPU + CPU re-encode + egress) is attributable per job; stitch-on-laptop hides it.

707 - Aligns with ADR-012 (GPU-native) + ADR-013 (GCP sole stack) + the 2026-06-20 deploy report; the truly-local profile preserves the ADR-009 privacy/offline self-host story.

708

709 **### Consequences**

710

711 - **Positive:** one core, provably identical across modes; CI proves the decoupling for **\$0** (stub + storage fakes + a profile-parity test); new compute targets (`byo_worker`, SkyPilot, Modal) slot into the enum without core changes; **cost becomes visible per job** (incl. stitch).

712 - **Cost/new surface:** a new `deployment+compute_target` config surface, a Cloud Run **dispatch shim**, a **containerized GPU+ffmpeg worker image**, a **cost-ledger migration**, and **edge auth** must all be built + maintained. Cross-GPU detection is **not** byte-identical ? parity asserted at the window level. The `byo_worker` tier carries a real ops/support burden (deferred, power-user only).

713 - **Risk-managed:** every core-touching change is default-OFF + golden/parity-gated + Launch-Reported; real-GPU runs are **owner-side** with a budget guard and a posterity `DEPLOY_REPORT` (`docs/COMPUTE_DECOUPLED_SERVING/runlogs/`).

714 - **Owner gates** ? the 5 gating ones LOCKED 2026-06-21 (project doc \$2 **D7?D12**): D7 cold-start = **scale-to-zero**; D8 pricebook = **versioned DB table**, USD-internal/INR-display; D9 edge-auth = **Cloudflare Access** (verify Cf-Access JWT, lock ingress); D10 = **free-tier data-for-reels trade** (adults-only, derived-data, counsel-gated to M9); D11 quality-floor = **Gemini-ON until the owned model clears the floor** + D12 the **data-flywheel harness** (capture?store?eval?goldenize). Plus **3 more locked at M0 sign-off** (2026-06-21, \$2 **D13?D15**): **D13** profile is user-switchable anytime (advertised feature; equivalent output, not bit-identical cross-GPU); **D14 NORTHSTAR** = mobile/cloud-native point-at-Google-Drive ? reel back in Drive next to the source (implies a new `gdrive-api OAuth` backend = M12; resolves the gdrive-in-cloud question); **D15** auto-detect SUGGESTS, the user CONFIRMS ? execution (esp. cloud spend) is never a surprise. Still open (non-gating): stitch co-located vs split; NVDEC vs libx264; the `<2GB` cap; the dir name.

715

7. assistant (2026-07-03T16:45:34.107Z)

Now let me examine the specific sourcing of D3 and D4 in the table, and cross-check against the template and ADR conventions.

Looking at **\$2 "Decisions locked"** in ALPHA_LAUNCH_READINESS.md (lines 29?35):

```
| Decision | Source / date |
|---|---|
| D3 | `data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 |
```

```
| **D4** | `backend.eval.serve_contrast` run, 2026-07-03 |
```

Compared to D1, D2, D5:

```
| Decision | Source / date |
|---|---|
| **D1** | `NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 |
| **D2** | `NEXT_STEPS.md` P0 UI-driven/shareable item; verified 2026-07-03 |
| **D5** | sibling `kheIsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 |
```

Now check PROJECT_DOC_TEMPLATE.md (lines 35?40):

```
...
```

2. Decisions locked

<Table ? capture answered questions so they are not relitigated.>

```
| # | Decision | Source / date | Implication |
|---|-----|-----|-----|
| D1 | ? | ? | ? |
|...
```

The template says "capture answered questions so they are not relitigated" but does not explicitly define what "locked" means or require owner sign-off/ADR ratification.

Now check DECISIONS.md for conventions. Looking at ADR-001 through ADR-014 (lines 1?715), I can see:

- Each ADR has a **Date**, **Status** (Accepted/Proposed/Ratified), and **Context/Decision/Consequences** structure.
- **ADR-008** (lines 388?469) explicitly says **"Status: Accepted (owner-ratified after an adversarial design panel)"** ? showing that ratified decisions carry owner sign-off explicitly stated in the status line.
- **ADR-014** (line 688) says **"Status: ? RATIFIED (owner, 2026-06-21 ? M0 design sign-off; build M1+.)"** ? again, owner ratification is **explicitly recorded**.

Checking DOC_STATUS.md (lines 1?98):

- §2 (lines 14?24) defines the **archival checklist** which includes: **"Flip Status ? DELIVERED / DONE + a completion note"** and **"verify claims against the code" ? no stale assertions; tag load-bearing facts `[verified]`."**
- Line 5 says the doc reflects **"real shipped state, never aspirational"** and should keep it **HONEST**.
- Line 12 says: **"Keep it HONEST: tag each claim `[verified]` (checked against code), `[researched]` (needs sign-off), or `[design]` (proposed)."**

The **critical evidence**: In ALPHA_LAUNCH_READINESS.md's TL;DR (lines 7?8), it uses **two honesty tags**:

- **`[verified]`** = checked against code or a live doc
- **`[analysis]`** = derived from a local command run

And line 13 explicitly says: **"This is not legal, financial, or launch approval."**

D3 and D4 are sourced **only** as **`[analysis]`** (local eval runs, 2026-07-03), whereas D1/D2/D5 are sourced as either direct doc refs or **`[verified]`** against a doc or config item.

Verdict Analysis:

1. **D3's source** ("local eval runs, 2026-07-03") is author-run analysis with no owner sign-off cited, no ADR.
2. **D4's source** ("backend.eval.serve_contrast run, 2026-07-03") is likewise author-run with no owner sign-off.
3. The repo's **PROJECT_DOC_TEMPLATE.md** does not mandate owner/ADR ratification for "Decisions locked" ? the template is silent on the gatekeeping rule.
4. **However,** DECISIONS.md's conventions (ADR-001?014) show that **ratified decisions** explicitly state owner sign-off in the Status line (e.g., "RATIFIED (owner, 2026-06-21)").
5. The repo's **honesty standard** (DOC_STATUS.md §2, ALPHA_LAUNCH_READINESS.md line 7) requires

tagging claims as `[verified]`, `[researched]`, or `[design]`, and the doc author used `[analysis]` to distinguish author-local runs from owner-verified or external-source facts.

6. D3 and D4 **are not tagged `[verified]` at the decision row level** ? the table header does not carry honesty tags, only the source cell text says "local eval runs" and a specific command name. A reader seeing "Decisions locked" will treat the table as owner-ratified unless the source cell explicitly signals otherwise.

****CONFIRMED****

****Evidence:**** The repo's convention (exemplified in ADR-008, ADR-014, and DOC_STATUS.md §2) is that locked/ratified decisions must state owner sign-off explicitly. D3 sources "local eval runs, 2026-07-03" (no owner named); D4 sources a "backend.eval.serve_contrast run, 2026-07-03" (no owner sign-off). The PROJECT_DOC_TEMPLATE.md is silent on gatekeeping, but the DECISIONS.md ADR log shows every ratified decision carries an explicit "RATIFIED (owner, date)" or equivalent status line. D3 and D4 lack this and are sourced only from the doc author's own analysis runs, not from an owner decision, ADR, or signed-off memo.

****Suggested fix:**** Rename §2 "Decisions locked" ? "Decisions draft" or "Premises for this plan", and retag D3/D4's source cells to *****[analysis]**** local eval runs, 2026-07-03" (matching the honesty-note convention used in the TL;DR) so a reader understands these are the author's evidence-based proposals, not yet owner-ratified. Then file a cross-repo issue to record owner's formal sign-off once reviewed.